Ai Assisted Coding

Assignment-7.5

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Batch:45

Task 1 (Mutable Default Argument – Func on Bug)

Task: Analyze given code where a mutable default argument causes

unexpected behavior. Use AI to fix it. # Bug: Mutable default

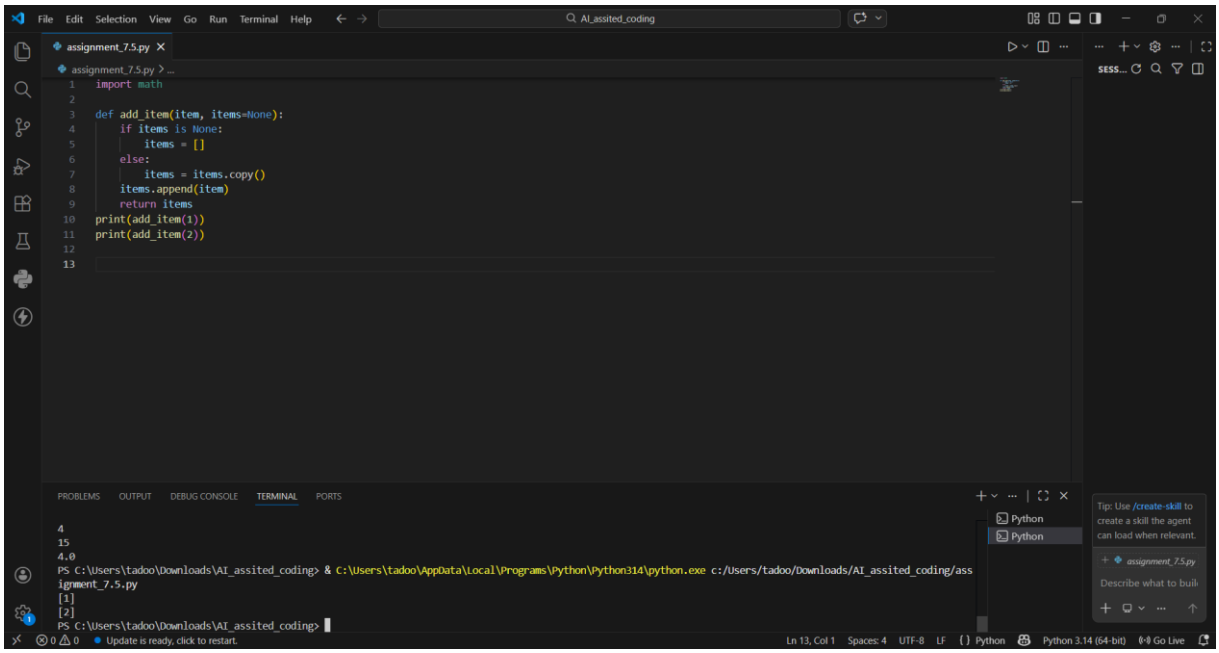
argument def add_item(item, items=[]):

items.append(item) return

items print(add_item(1))

print(add_item(2))

Expected Output: Corrected func on avoids shared list bug.



Task 2 (Floa ng-Point Precision Error)

Task: Analyze given code where floa ng-point comparison fails.

Use AI to correct with tolerance. #

Bug: Floa ng point precision issue

def check_sum(): return (0.1 + 0.2)

== 0.3 print(check_sum())

Expected Output: Corrected func on

```
def check_sum():
    return (0.1 + 0.2) == 0.3
    return math.isclose(0.1 + 0.2, 0.3,
print(check_sum())
```

Task 3 (Recursion Error – Missing Base Case)

Task: Analyze given code where recursion runs infinitely due to

missing base case. Use AI to fix.

Bug: No base case

def countdown(n):

print(n)

return countdown(n-1) countdown(5)

Expected Output : Correct recursion with stopping condi on.

```
def countdown(n):
    if n < 0:
        return
    print(n)
    return countdown(n-1)
    if n > 0:
        return countdown(n-1)
countdown(5)
```

Task 4 (Dic onary Key Error)

Task: Analyze given code where a missing dic onary key causes

error. Use AI to fix it.

Bug: Accessing non-exis ng key

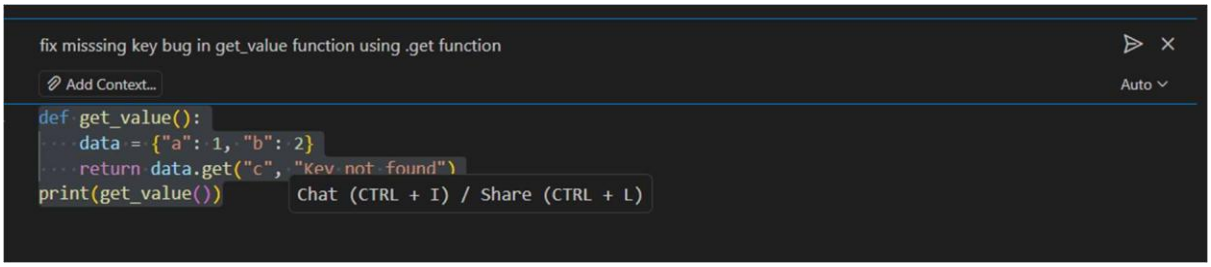
def get_value(): data

= {"a": 1, "b": 2}

return data["c"]

print(get_value())

Expected Output: Corrected with .get() or error handling.



Task 5 (Infinite Loop – Wrong Condi on)

Task: Analyze given code where loop never ends. Use AI to detect

and fix it.

Bug: Infinite loop

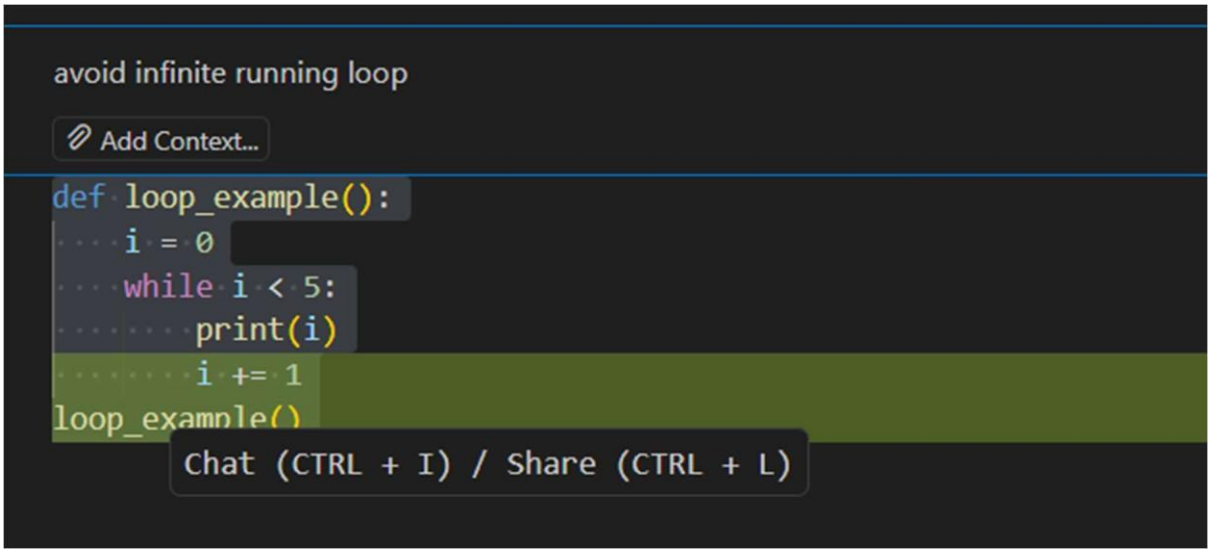
def loop_example():

i = 0 while

i < 5:

print(i)

Expected Output: Corrected loop increments i.



Task 6 (Unpacking Error – Wrong Variables)

Task: Analyze given code where tuple unpacking fails. Use AI to

fix it.

Bug: Wrong unpacking

a, b = (1, 2, 3)

Expected Output: Correct unpacking or using _ for extra values`

```
a, b, c = 1, 2, 3
```

Task 7 (Mixed Indenta on – Tabs vs Spaces) Task:

Analyze given code where mixed indenta on breaks execu

on. Use AI to fix it. # Bug: Mixed indenta on

def func():

x = 5 y =

10 return

x+y

Expected Output : Consistent indenta on applied.

```
Modify selected code
Add Context...
def func():
x = 5
y = 10
return x + y
    x = 5
    y = 10
    return x + y

print(func())
```

Task: Analyze given code with incorrect import. Use AI to fix.

Bug: Wrong import import

maths print(maths.sqrt(16))

Expected Output: Corrected to import math

```
import maths
print(maths.sqrt(16))
print(math.sqrt(16))
```